BITE401P — Embedded Systems and IoT

Project Review – 1

Team Size: 3 Members

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I. Title of the Project

Cloud-Based Carbon Intelligence System (CCIS): A Sensor-Fusion Embedded Framework for Real-Time Household Carbon Footprint Estimation and Credit Scoring

II. Field / Area of Project

PRIMARY DOMAIN	APPLICATION AREA	ENABLING TECHNOLOGY
Internet of Things (IoT) & Embedded Systems	Environmental Sustainability / Green Computing	Cloud Computing & Machine Learning (Edge–Cloud Inference)

III. Prior Patents or Publications

Reference	Summary of Existing Work	Limitation / Gap	How Our Work Differs
Olatomiwa et al., "IoT-Based Visualisation Platforms for Tracking Household Carbon Footprints," Sustainability (MDPI), 2023	Reviews IoT sensor platforms (e.g., Carbon Track) that monitor appliance energy, water, temperature and humidity, and push data to a cloud dashboard for visualization.	Focused on visualization/monitoring only; no occupancy-aware inference, no credit-scoring or actionable incentive layer; largely a survey, not a deployed low-cost prototype.	Adds multi-sensor fusion (power + gas + occupancy), an inference/scoring engine, and a physical low-cost hardware prototype.
US 2021/0224819 A1 – "Carbon Footprint Tracker" (Google Patents)	Software system that scans e-commerce/point-of-sale transactions on a computing device to estimate the carbon footprint of purchases and suggest lower-emission alternatives.	Purely transaction/software based; does not use physical embedded sensors or real-time environmental/energy data; no hardware component.	Our system uses real embedded hardware (current, gas, occupancy sensors) rather than transaction logs, capturing actual physical energy usage.
US 2009/0210295 A1 – "System and Method for Enabling Carbon Credit Rewards for Select Activities"	Assigns carbon credits to specific user activities (e.g., driving, household energy use) monitored via an onboard/household monitor, rewarding reduction behavior.	Generic activity-monitor concept from 2009; predates modern low-cost IoT microcontrollers (ESP32), cloud ML pipelines, and lacks a concrete sensor-fusion or edge-inference architecture.	Implements the credit-reward concept using a modern, cheap ESP32-based multi-sensor node with a cloud ML inference layer for score prediction.
US 2025/0166067 A1 – "Carbon Emissions Tracking and Trading Using Blockchain"	Blockchain-based smart-contract platform for registering emitters and automating carbon-credit trading between companies/entities.	Targets industrial/corporate emitters and financial trading infrastructure; not designed for individual household deployment or embedded sensing.	Scoped to individual households/small offices with a simulated local credit-scoring layer instead of an industrial trading exchange.

Prior-art search conducted via Google Patents and academic literature (MDPI, IEEE-indexed sources). A formal patentability search via the University Central Library patent-search facility is planned prior to any filing, as recommended in the course instructions.

IV. Abstract

Rising household energy consumption is a significant but often invisible contributor to national carbon emissions, and most existing IoT carbon-tracking systems either measure ambient CO_2 levels which poorly represent an individual household's actual footprint or rely purely on financial transaction data with no physical sensing at all. This project proposes a Cloud-Based Carbon Intelligence System (CCIS), a low-cost embedded IoT device that fuses appliance-level current draw, ambient gas concentration, temperature/humidity, and

room occupancy data to compute a household's real-time carbon emission footprint. The embedded node (built on an ESP32 microcontroller) performs lightweight edge pre-processing and streams sensor data to a cloud platform, where a machine-learning-based inference layer estimates the effective carbon footprint using regional grid emission factors and occupancy-adjusted usage patterns. The cloud engine further predicts monthly emission trends and generates a gamified Carbon Credit Score, encouraging users to reduce idle and wasteful energy consumption through actionable dashboard recommendations and optional automated appliance cut-off. Unlike prior work that is limited to either pure visualization dashboards or purely software/transaction-based tracking, this system integrates physical multi-sensor fusion, edge-cloud machine learning inference, and an incentive-driven scoring mechanism into a single low-cost (under 2000), field-deployable prototype, making it suitable for both practical household adoption and academic publication as a novel sensor-fusion carbon-intelligence framework.

V. Objectives of the Project

1. **Design and deploy a low-cost multi-sensor embedded hardware node** (ESP32-based) capable of simultaneously acquiring appliance current draw, ambient gas concentration, temperature/humidity, and occupancy data in real time.
2. **Develop a cloud-based data ingestion and storage pipeline** to reliably transmit, store, and timestamp sensor data streams using MQTT/HTTPS protocols for downstream processing.
3. **Build a machine-learning-based carbon inference engine** that converts raw multi-sensor readings into an accurate, occupancy-adjusted household carbon emission estimate using regional grid emission factors.
4. **Design and implement a gamified Carbon Credit Scoring mechanism** that translates predicted emissions into a comparative score and generates actionable, personalized recommendations for emission reduction.
5. **Develop a cloud-hosted dashboard/web interface** for real-time visualization of footprint trends, historical comparisons, and credit score progress.
6. **Validate system accuracy and cost-effectiveness** by benchmarking predicted emissions against manually calculated ground-truth values and standard appliance energy ratings.

VI. System Architecture

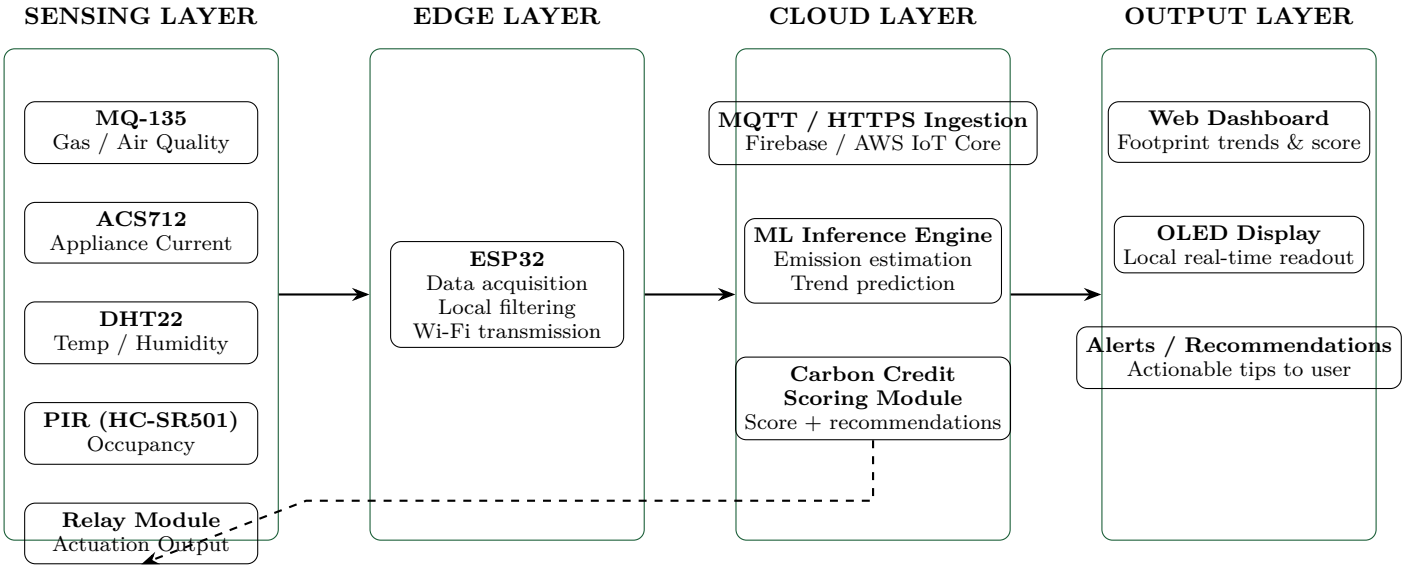


Fig. 1 End-to-end system architecture: sensing layer edge processing (ESP32) cloud ingestion ML inference & credit scoring dashboard/output & actuation feedback.

VII. Hardware and Software Details with Specifications

A. Hardware Components

Component	Specification	Purpose in System
Microcontroller	ESP32 DevKit V1 Dual-core 240 MHz, Wi-Fi + BLE, 34 GPIOs, 12-bit ADC	Central processing unit; edge pre-processing and cloud connectivity
Gas Sensor	MQ-135 Detects CO_2, NH_3, NOx, alcohol, benzene; analog output, 5V operation	Proxy indicator of indoor air quality / combustion-related emissions
Current Sensor	ACS712 (20A variant) Hall-effect based, analog output, ś185 mV/A sensitivity	Measures real-time appliance current draw to compute energy consumption (kWh)
Temperature & Humidity Sensor	DHT22 Range: -40–80řC (ś0.5řC), 0–100% RH (ś2%)	Contextualizes HVAC-related energy usage and comfort-driven consumption
Occupancy Sensor	HC-SR501 PIR Module 7m detection range, adjustable delay	Detects room occupancy to differentiate active usage from idle/standby drain
Relay Module	5V 1-Channel Relay (10A/250VAC rated)	Enables automated cut-off of idle high-drain appliances (actuation output)
Display (optional)	0.96" I2C OLED, 128Œ64 px, SSD1306 driver	Local real-time display of footprint/score without needing the dashboard
Power Supply	5V/2A USB adapter with buck converter regulation	Powers ESP32 and peripheral sensor modules

B. Software & Cloud Stack

Layer	Tool / Technology	Purpose
Firmware	Arduino IDE / ESP-IDF (C/C++)	Sensor data acquisition, edge filtering, and Wi-Fi transmission logic
Communication Protocol	MQTT (via broker) / HTTPS REST	Lightweight, reliable transmission of sensor payloads to the cloud
Cloud Platform	Firebase Realtime Database / AWS IoT Core (free tier)	Ingestion, timestamping, and storage of streamed sensor data
Inference Engine	Python (scikit-learn) Linear Regression / Decision Tree model, deployed as a cloud function	Converts raw sensor fusion data into carbon footprint estimates and trend predictions
Dashboard	Firebase Hosting + HTML/CSS/JavaScript (Chart.js for visualization)	User-facing web dashboard showing footprint trends and credit score
Development & Version Control	VS Code, Git/GitHub	Collaborative firmware and dashboard development across the 3-member team